A PRO-ISOMORPHISM OF INVERSE SYSTEMS IN A TRIANGULATED CATEGORY INDUCES AN ISOMORPHISM OF THEIR DERIVED LIMITS

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Allowed standard prerequisites (editorial).

- Pro-objects and pro-isomorphisms of inverse systems. the hypothesis at statement lines 23173-23174. Fixes the weakening of an isomorphism that the argument must convert into an isomorphism of limits.
- The derived limit of an inverse system in a triangulated category. used throughout the counted proof span. Is the construction whose behaviour under pro-isomorphism is being established.

Source reference. Original lemma 0H9J: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21529–21529. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 23149: sources/stacks-more-algebra/pinned-original-source.tex, source lines 23149–23149. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 23073: sources/stacks-more-algebra/pinned-original-source.tex, source lines 23073–23073. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $\mathcal{D}$ be a triangulated category. Let (K_n) and (M_n) be inverse systems of objects of $\mathcal{D}$ with derived limits K and M. Let $a : (K_n) \to (M_n)$ be a pro-isomorphism of pro-objects. Then a can be used to produce a (non-canonical) isomorphism $K \to M$.*

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Proof. We obtain an arrow $K \to M$ fitting into a morphism

$$\begin{array}{ccccc} K & \longrightarrow & \prod K_n & \longrightarrow & \prod K_n \\ \downarrow & & \downarrow{\scriptstyle a'} & & \downarrow{\scriptstyle a''} \\ M & \longrightarrow & \prod M_n & \longrightarrow & \prod M_n \end{array}$$

of defining distinguished triangles by Derived Categories, Remark 0H9J. Thus, for every object L of $\mathcal{D}$ we obtain a map of short exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & R^1 \lim \mathrm{Hom}_{\mathcal{D}}(L, K_n[-1]) & \longrightarrow & \mathrm{Hom}_{\mathcal{D}}(L, K) & \longrightarrow & \lim \mathrm{Hom}_{\mathcal{D}}(L, K_n) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & R^1 \lim \mathrm{Hom}_{\mathcal{D}}(L, M_n[-1]) & \longrightarrow & \mathrm{Hom}_{\mathcal{D}}(L, M) & \longrightarrow & \lim \mathrm{Hom}_{\mathcal{D}}(L, M_n) & \longrightarrow & 0 \end{array}$$

see Lemma 23149 and its proof. By Lemma 23073 the left and right vertical arrows are isomorphisms[1]. Thus the middle arrow is an isomorphism. By the Yoneda lemma we see that the map $K \to M$ is an isomorphism. □

[1]If the pro-isomorphism a is given by maps $a_n : K_{m_n} \to M_n$ the we obtain a pro-isomorphism on Hom using the corresponding maps $\mathrm{Hom}_{\mathcal{D}}(L, K_{m_n}) \to \mathrm{Hom}_{\mathcal{D}}(L, M_n)$. Thus the construction of the arrows on lim and R^1 lim in the proof of Lemma 23073 agrees with the construction of a' and a'' in Derived Categories, Remark 0H9J.